Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Takes out a factor of 2, or obtains a = 2 by equating coefficients. Note $2(x^2$ − 'anything') or $2(x$ − 'anything'$)^2$ scores M1	AO1.1a	M1	$2\left(x^2-\frac{5x}{2}\right)+k$ $2\left(x-\frac{5}{4}\right)^2+k-\frac{25}{8}$
	Expresses as $\left(x-\frac{5}{4}\right)^2$ or obtains $b=\frac{5}{4}$ by equating coefficients	AO1.1a	M1	
	Obtains correct expression. If using equating coefficients **must** be put back in given form required	AO1.1b	A1	

	Marking Instructions	AO	Marks	Typical Solution
(b)	Selects a method using completed square form (recognition that vertex occurs when x = 'their' b), discriminant (any use of $b^2 - 4ac$ seen) or calculus (finds y coordinate of stationary point).	AO3.1a	M1	Typical solution 1 When $x=\frac{5}{4}$ $\left(x-\frac{5}{4}\right)^2=0$ $\left(k-\frac{25}{8}\right)>3$ $k>\frac{49}{8}$
	Forms an appropriate correct inequality for their chosen method. (first time inequality sign seen in each typical solution scores A1)	AO1.1b	A1	
	Obtains $k>\frac{49}{8}$ (ACF)(OE)	AO1.1b	A1	
				Typical solution 2 $2x^2 - 5x + k = 3$ $2x^2 - 5x + k - 3 = 0$ $5^2 - 4 \times 2 \times (k - 3) < 0$

			$k > \frac{49}{8}$
			Typical solution 3 $\frac{dy}{dx} = 4x - 5$ At stationary point $\frac{dy}{dx} = 0$ $4x - 5 = 0$ $x = \frac{5}{4}$ when $x = \frac{5}{4}$ $y = k - \frac{25}{8}$ $k - \frac{25}{8} > 3$ $k > \frac{49}{8}$
			Total 6 marks

Q2.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO2.5	B1	A $\Leftarrow$ B
			Total 1 mark

Q3.

Marking Instructions	AO	Marks	Typical Solution
Obtains $\frac{dy}{dx}$ for both the given curves – at least one term must be correct for each curve	AO3.1a	M1	$\frac{dy}{dx} = 6x^2 + 12x - 12$ $\frac{dy}{dx} = 60 - 12x$
States both derivatives correctly	AO1.1b	A1	
Translates problem into an inequality	AO3.1a	M1	Chris's claim is **incorrect** when
States a correct quadratic inequality	AO1.1b	A1	$6x^2 + 12x - 12 \leq 60 - 12x$ $2x^2 + 8x - 24 \leq 0$

FT from an incorrect $\frac{dy}{dx}$ provided both M1 marks have been awarded			$x^2 + 4x - 12 \leq 0$ $(x + 6)(x - 2) \leq 0$ Critical values are $x = -6$ and 2 region: $x < -6$ / $-6 < x < 2$ / $x \leq 2$ sign: + / − / +
Determines a solution to 'their' inequality	AO1.1a	M1	$-6 \leq x \leq 2$ Chris's claim is incorrect for values of x in the range $-6 \leq x \leq 2$, so he is wrong
Obtains correct range of values for x Must be correctly written with both inequality signs correct	AO1.1b	A1	
Interprets final solution in context of the original question, must refer to Chris's claim	AO3.2a	R1	
			Total 7 marks

Q4.

(a) $(k - 2)^2 - 4 \times (2k - 7)(k - 3)$

discriminant – condone one slip – condone omission of brackets

M1

$k^2 - 4k + 4 - 4(2k^2 - 6k - 7k + 21)$

A1

" their " $-7k^2 + 48k - 80 \geq 0$

real roots condition; f(k) ≥ 0 must appear before final line

B1

$7k^2 - 48k + 80 \leq 0$

AG *(all working correct with no missing brackets etc)*

A1cso

4

(b) $7k^2 - 48k + 80 = (7k - 20)(k - 4)$

correct factors

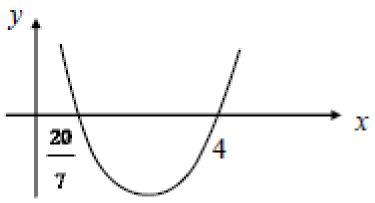
(or roots unsimplified) $\dfrac{48 \pm \sqrt{64}}{14}$

M1

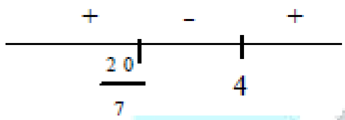
critical values are 4 and $\dfrac{20}{7}$

accept $\dfrac{56}{14}$, $\dfrac{40}{14}$ *etc here*

A1



sketch or sign diagram including values



M1

$\dfrac{20}{7} \leq k \leq 4$

Take their final line as their answer

fractions must be simplified here

A1cao

4

[8]

Q5.

(a) $(2x - 3 = x)$

$x = 3$

B1

$2x - 3 = -x$

or $-(2x - 3) = x$ *or* $-2x + 3 = x$

M1

$x = 1$

A1

3

(b) $(|2x - 3| \geq |x|)$

No ISW in part(b), mark their final line as their answer.

$x \leq 1$

Or $1 \geq x$

B1

$x \geq 3$

Or $3 \leq x$

Or "$x \leq 1$ or $x \geq 3$" for B1 B1

B1

2

[5]

Q6.

(a) Sides are x and $x + 4$

$$\left.\begin{aligned} &\Rightarrow x + x + x + 4 + x + 4 > 30 \\ &\text{or } \ 2x + 2x + 8 > 30 \\ &\text{or } \ 2(2x + 4) > 30 \\ &\text{or } \ 4x + 8 > 30 \end{aligned}\right\}$$

must see this line OE

$(\Rightarrow 4x > 22)$

$\Rightarrow 2x > 11$

***AG** (be convinced) condone $11 < 2x$*

B1

(b) $x(x + 4) < 96$

must see this line OE

$\Rightarrow x^2 + 4x - 96 < 0$

AG

B1

1

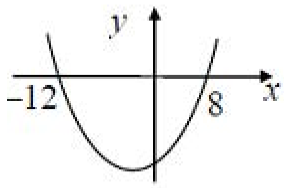
(c) $(x + 12)(x - 8)$

correct factors or correct quadratic equation formula

M1

Critical values 8, −12

A1



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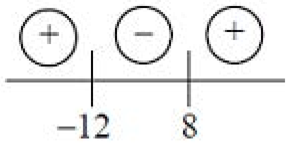


sketch or sign diagram

M1

⇒ $-12 < x < 8$;

accept $x < 8$ ***AND*** $x > -12$

*but **not*** $x < 8$ ***OR*** $x > -12$

nor $x < 8, x > -12$

A1cso

(d) $5\frac{1}{2} < x < 8$

B1

1

[7]

Q7.

(a)

stretch I
SF 4 II
in y-direction III

translate $\begin{pmatrix} e \\ 0 \end{pmatrix}$

either order

I + (II or III)

M1A1

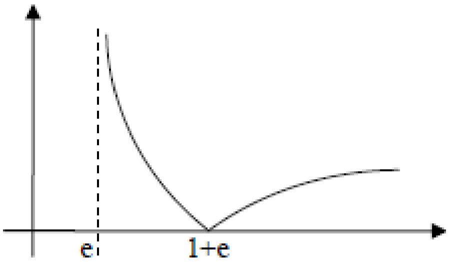
E1

accept 'e in positive x-direction'

B1

4

(b)



mod graph, in 2 connected sections, both in the first quadrant, touching x-axis

M1

curve touches x-axis at 1 + e (or 3.7 or better), and labelled (ignore scale)

A1

correct curvature, including at their 1 + e, approx. asymptote at $x = e$

A1

3

(c) (i) $| 4\ln(x - e) | = 4$

$4\ln(x - e) = 4$ $4\ln(x - e) = -4$ or better

must see 2 equations, condone omission of brackets

M1

(x =) 2e do not ISW

accept values of AWRT 5.42, 5.43, 5.44

A1

(x =) e + e^{-1} or (x =) e + $\frac{1}{e}$ do not ISW

accept values of AWRT 3.08, 3.09

if M0 then x = 2e with or without working scores SC1

A1

3

(ii) $x \geq 2e$

accept values of AWRT 5.42, 5.43, 5.44

B1

$e < x \leq e + \frac{1}{e}$

accept values of AWRT 2.72, 3.08, 3.09

if B2 not earned, then SC1for any of

$e \le x \le e + \frac{1}{e}$, $e < x < e + \frac{1}{e}$, $e \le x < e + \frac{1}{e}$

B2

3

[13]

Q8.

(a) (i)

$$\left.\begin{array}{l}(\text{Increasing} \Rightarrow)\ \dfrac{dy}{dx} > 0 \\ 20x - 6x^2 - 16 > 0\end{array}\right\}$$ either

correct interpretation of y increasing

M1

$\Rightarrow 6x^2 - 20x + 16 < 0$ *or* (2) $(10x - 3x^2 - 8) > 0$

must see at least one of these steps before final answer for A1

$\Rightarrow 3x^2 - 10x + 8 < 0$

*CSO **AG** no errors in working*

A1

2

(ii) $(3x - 4)(x - 2)$

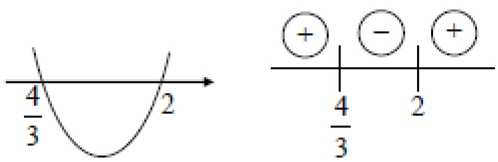
correct factors or correct use of quadratic equation formula as far as $\frac{10 \pm \sqrt{4}}{6}$

M1

CVs are $\frac{4}{3}$ and 2

condone $\frac{8}{6}$ and $\frac{12}{6}$ *here but not in final line*

A1



sketch or sign diagram

M1

$\frac{4}{3} < x < 2$

or $2 > x > \frac{4}{3}$

accept $x < 2$ ***AND*** $x > \frac{4}{3}$

but ***not*** $x < 2$ ***OR*** $x > \frac{4}{3}$

nor $x < 2,\ x > \frac{4}{3}$

A1

Mark their final line as their answer

4

(b) (i) $x = 2$; $\left(\frac{dy}{dx} =\right)$ 40 − 24 − 16

sub $x = 2$ *into* $\frac{dy}{dx}$ *and simplify terms*

M1

$\frac{dy}{dx} = 0 \Rightarrow$ tangent at P is parallel to the x-axis

must be all correct working plus statement

A1

2

(ii) $x = 3$; $\frac{dy}{dx} = 20 \times 3 - 6 \times 3^2 - 16$

must attempt to sub $x = 3$ *into* $\frac{dy}{dx}$

M1

(= 60 − 54 − 16) = − 10

A1

Gradient of normal = $\frac{1}{10}$

$\frac{-1}{\text{"their } -10\text{"}}$

A1 √

Normal: $(y - -1) =$ 'theirgrad'$(x - 3)$

normal attempted with correct coordinates

used and gradient obtained from their $\frac{dy}{dx}$ *value*

m1

$y + 1 = \frac{1}{10}(x - 3)$

any correct form, eg 10 y = x − 13 but must simplify − − to +

A1

(Equation of tangent at P is) y = 3

B1

x = 43

CSO; ⇒ R (43, 3)

A1

7

[15]

Q9.

(a) (i) (–) $(x + 5)^2$

q = 5; condone $(–x –5)^2$

M1

$29 – (x + 5)^2$

p = 29 and q = 5

A1

2

(ii) x = –5 is line of symmetry

FT x = – 'their q' or correct

B1√

1

(b) (i) $4 – 10x – x^2 = k(4x – 13)$

$\Rightarrow x^2 + 4kx + 10x - 13k - 4 = 0$

Must see both these lines OE

$\Rightarrow x^2 + 2(2k + 5)x - (13k + 4) = 0$

AG all correct working and = 0

B1

1

(ii) 2 distinct roots $\Rightarrow b^2 - 4ac > 0$

stated or used (must be > 0)

B1

Discriminant = $4(2k + 5)^2 + 4(13k + 4)$

condone one slip (may be within formula)

M1

$4(4k^2 + 20k + 25 + 13k + 4) > 0$

or $16k^2 + 132k + 116 > 0$

$\Rightarrow 4k^2 + 33k + 29 > 0$

AG > 0 must appear before final line

A1

3

(iii) $(4k + 29)(k + 1)$

correct factors or correct unsimplified quadratic

equation formula $\dfrac{-33 \pm \sqrt{33^2 - 4 \times 4 \times 29}}{8}$

M1

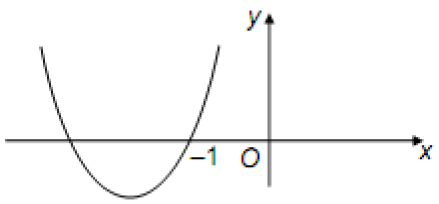
$k = -\dfrac{29}{4}$, $k = -1$

condone $k = -\dfrac{58}{8}$, *7.25 etc but not left with*

square roots etc as above

A1

$-\dfrac{29}{4}$



sketch or sign diagram including values

M1

$k < -\frac{29}{4}$, $k > -1$

*condone use of **OR** but not **AND***

A1

Take their final line as their answer

4

[11]

Q10.

(a) $8 - 6x > 5 - 4x - 8$

multiplying out correctly and > sign used

M1

$11 > 2x$

$x < 5\frac{1}{2}$ $\left(\text{or } x < \frac{11}{2}\right)$

accept 5.5 > x OE

A1cso

2

(b) $2x^2 + 5x - 12 \geq 0$

$(x + 4)(2x - 3)$

correct factors

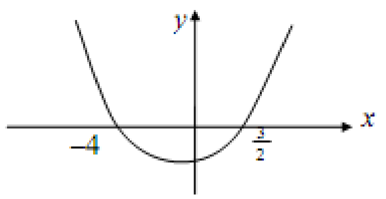
(or roots unsimplified) $\frac{-5 \pm \sqrt{121}}{4}$

M1

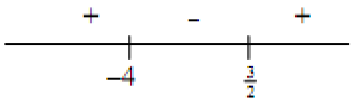
Critical values are $-4 \text{ and } \frac{3}{2}$

both CVs correct; condone $\frac{6}{4}, -\frac{16}{4}$ *etc here but must be single fractions*

A1



sketch or sign diagram including values



M1

$x \leq -4, \; x \geq \frac{3}{2}$

take their final line as their answer

fractions must be simplified
condone use of OR but not AND

A1 **4**

[6]

Q11.

(a) (i) $x = \pm 2$ or $y = \pm 6$ or $(x - 2)^2 + (y + 6)^2$

M1

C 2, –6

correct

A1 **2**

(ii) $r^2 = 4 + 36 - 15$

(RHS =) their $(-2)^2$ + their $(6)^2 - 15$

M1

$\Rightarrow r = 5$

Not ±5 or $\sqrt{25}$

A1 **2**

(b) explaining why $|y_C| > r$; 6 > 5

Comparison of y_C and r, eg –6 + 5 = –1
or indicated on diagram

E1

full convincing argument, but must have
correct y_C and r

Eg "highest point is at y = –1" scores E2
E1: showing no real solutions when y = 0

+E1 stating centre or any point below x-axis

E1 — **2**

(c) (i) $PC^2 = 5 - 2^2 + k + 6^2$

ft their C cords

$= 9 + k^2 + 12k + 36$

and attempt to multiply out

M1

$PC^2 = k^2 + 12k + 45$

AG CSO (must see PC^2 = at least once)

A1 — **2**

(ii) $\left.\begin{array}{l} PC > r \Rightarrow PC^2 > 25 \\ \Rightarrow k^2 + 12k + 20 > 0 \end{array}\right\}$

AG Condone $\left.\begin{array}{l} k^2 + 12k + 45 > 25 \\ \Rightarrow k^2 + 12k + 20 > 0 \end{array}\right\}$

B1 — **1**

(iii) $k + 2 \quad k + 10$

***Correct** factors or **correct** use of formula*

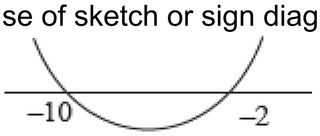
M1

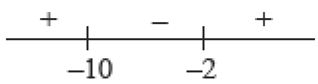
$k = -2$, $k = -10$ are critical values

May score M1, A1 for correct critical values seen as part of incorrect final answer with or without working.

A1

Use of sketch or sign diagram:





If previous A1 earned, sign diagram or sketch must be correct for M1, otherwise

M1 may be earned for an attempt at the sketch or sign diagram using their critical values.

M1

$\Rightarrow$ $k > -2$, $k < -10$

$k \geq -2$, $k \leq -10$ loses final A mark

A1

Condone $k > -2$ OR $k < -10$ for full marks but not AND instead of OR
Take final line as their answer

Answer only of $k > -2$, $k > -10$ etc scores M1, A1, M0 since the critical values are evident.

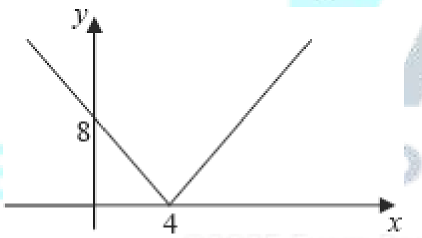
Answer only of $k > 2$, $k < -10$ etc scores M0, M0 since the critical values are not both correct.

4

[13]

Q12.

(a)



Modulus graph V shape in 1st quad going into 2nd quad, touching x-axis. Must cross y-axis
Condone not ruled

M1

4 and 8 labelled

A1

2

(b) $x = 2$

One correct answer

B1

$x = 6$

Second correct answer and no extras
Condone answers shown on the graph, if clearly indicated

B1

2

(c) $x > 6$

One correct answer

B1

$x < 2$

Second correct answer and no extras and no further incorrect statement eg
$6 < x < 2$ or $2 < x > 6$
SC $x \geq 6$, $x \leq 2$ scores B1

B1

2

[6]

Q13.

(a) (i) $2(x - 5)^2$

$p = 5$

B1

$+3$

$q = 3$

B1

2

(ii) Stating both $(x - 5)^2 \geq 0$ and $3 > 0$

FT their p & q, but must have $q > 0$

M1

$\Rightarrow$ $2x^2 - 20x + 53 > 0$ or $2(x - 5)^2 + 3 > 0$

$\Rightarrow$ $2x^2 - 20x + 53 = 0$ has no real roots

Must have statement and correct p & q.

A1cso

2

(b) (i) $b^2 - 4ac = (k + 1)^2 - 4k(2k - 1)$

Condone one slip (including x is one slip)

M1

$= -7k^2 + 6k + 1$

Condone recovery from missing brackets

A1

real roots $\Rightarrow$ $b^2 - 4ac \geq 0$

Their discriminant ≥ 0 (in terms of k)

$-7k^2 + 6k + 1 \geq 0$

Need not be simplified & may earn earlier

B1ft

$\Rightarrow$ $7k^2 - 6k - 1 \leq 0$

AG (must see sign change)

A1cso

4

(ii) $(7k + 1)(k - 1)$

***Correct** factors or **correct** use of formula*

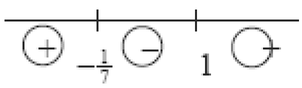
M1

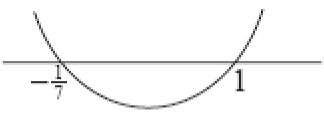
Critical values $k = 1, -\frac{1}{7}$

May score M1, A1 for correct critical values seen as part of incorrect final answer with or without working.

A1

Use of sign diagram or sketch





If previous A1 earned, sign diagram or sketch must be correct for M1

M1

*Otherwise M1 may be earned for an attempt at the sketch or sign diagram using **their** critical values.*

$-\frac{1}{7} \le k \le 1$

A1

Full marks for correct answer NMS

Condone $-\frac{2}{14}$ throughout

Condone $k \ge -\frac{1}{7}$ AND $k \le 1$ for full marks

Take their final line as their answer.

$\left(-\frac{1}{7} < k < 1\right)$, $\left(k \ge -\frac{1}{7}\ \mathbf{OR}\ k \le 1\right)$,

$\left(k \ge -\frac{1}{7}, k \le 1\right)$ scores M1A1M1A0

Answer only of $k < -\frac{1}{7},\ \ k < 1$ etc

scores M1, A1, M0 since the critical values are evident.

Answer only of $\frac{1}{7} \le k \le 1$ etc

scores M0, M0 since the critical values are not both correct.

4

[12]

Q14.

(a) $(x - 1)(2x - 3)$

$(1 - x)(3 - 2x)$ or $2(x - 1)(x - 1.5)$ etc

B1

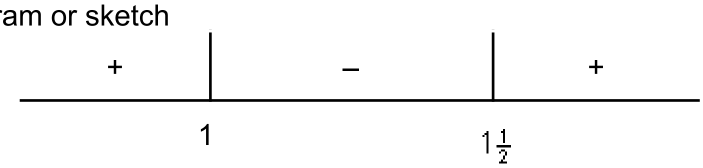
1

(b) Critical values are 1, $1\frac{1}{2}$

Correct or ft their factors from (a)

B1ft

Sign diagram or sketch



M1

$\Rightarrow$ $1 < x < 1\frac{1}{2}$

Full marks for correct inequality without working

A1

3

[4]

Q15.

(a) $\{S_{40} =\}$ $\frac{40}{2}[2a + (40-1)d]$

M1

$20(2a + 39d) = 1250$

A1

$\{25^{th}$ term $=\}$ $a + (25 - 1)d$

M1

$a + 24d = 38$

A1

Dep on both previous two Ms. Solving two equations in a and d simultaneously

m1

$18d = 27 \Rightarrow d = 1.5$

AG Be convinced

SC Using the given answer for d: mark out of a maximum of 4/6 as M1A1M1A1 {conclusion also needed in last A mark} (m0A0)

A1cso

6

(b) $a = 38 - 24 \times 1.5$

$= 2$

PI if using $a = 2$ ***in (b)***

If using eg $a = 38$ award this M mark at stage: no. of terms $\frac{100-38}{1.5} + 1 + 24$

M1

$a + (n - 1)\ 1.5 < 100$

M1

$n < \frac{100 - a}{1.5} + 1$

$n < 66.333\ .....$

$\Rightarrow$ number of terms < 100 is 66

NMS mark as B3 for 66 else B0

A1

3

[9]

Q16.

(a) $k(x^2 + 3) = 2x + 2$

$\Rightarrow kx^2 - 2x + 3k - 2 = 0$

AG OE all terms on one side and = 0

B1

1

(b) (i) Discriminant = $(-2)^2 - 4k(3k - 2)$

Condone one slip (including x is one slip)

Condone 2^2 or 4 as first term

M1

$= 4 - 12k^2 + 8k$

condone recovery from missing brackets

A1

Two distinct real roots $\Rightarrow b^2 - 4ac > 0$

$4 - 12k^2 + 8k > 0$

"their discriminant in terms of k" > 0

Not simply the statement $b^2 - 4ac > 0$

B1ft

$\Rightarrow 12k^2 - 8k - 4 < 0$

Change from > 0 to < 0 and divide by 4

$\Rightarrow 3k^2 - 2k - 1 < 0$

AG CSO

A1

4

(ii) $(3k + 1)(k - 1)$

__Correct__ factors or __correct__ use of formula

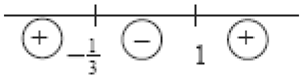
M1

Critical values 1 and $-\frac{1}{3}$

May score M1, A1 for correct critical values seen as part of incorrect final answer with or without working

A1

Use of sign diagram or sketch



If previous A1 earned, sign diagram or sketch must be correct for M1

M1

Otherwise, M1 may be earned for an attempt at the sketch or sign diagram using their critical values.

$\Rightarrow -\frac{1}{3} < k < 1$ or $1 > k > -\frac{1}{3}$

Full marks for correct final answer with or without working
≤ loses final A mark

A1

condone $-\frac{1}{3} < k$ AND $k < 1$ for full marks but not OR or "," instead of AND

Answer only of $1 < k < -\frac{1}{3}$ or $k < -\frac{1}{3};\ k < 1$ *etc scores M1, A1, M0 since the correct critical values are evident*

Answer only of $\frac{1}{3} < k < 1$ *etc where critical values are not both correct gets M0, M0*

[9]

Q17.

(a) $x^2 + 7 = k(3x + 1) \Rightarrow x^2 - 3kx + 7 - k = 0$

AG

B1

1

(b) $b^2 - 4ac = (-3k)^2 - 4(7 - k)$

Clear attempt at $b^2 - 4ac$
Condone slip in one term of expression

M1

(2 distinct roots when) $b^2 - 4ac > 0$

Must involve k

B1

$9k^2 + 4k - 28 > 0$

CSO; AG

A1

3

(c) $(9k - 14)(k + 2)$

Factors or formula correct unsimplified

M1

Critical points -2 and $\frac{14}{9}$

A1

Sketch $\cup$ or sign diagram correct

+ve	*–ve*	*+ve*
	-2	$\frac{14}{9}$

M1

$k < -2,\ k > \frac{14}{9}$

A1

4

[8]

Q18.

(a) $b^2 - 4ac = 16k^2 - 36(k + 1)$

Condone one slip

M1

Real roots: discriminant ≥ 0

B1

$\Rightarrow$ $16k^2 - 36k - 36 \geq 0$

$\Rightarrow$ $4k^2 - 9k - 9 \geq 0$

AG (watch signs)

A1

3

(b) $(4k + 3)(k - 3)$

Or correct use of formula (unsimplified)

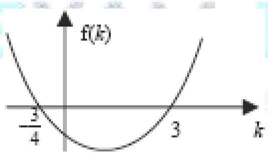
M1

critical points $(k =)$ $-\frac{3}{4}$, 3

Not in a form involving surds

Values may be seen in inequalities etc

A1



sketch

Or sign diagram

M1

$k \geq 3$, $k \leq -\frac{3}{4}$

NMS full marks

Condone use of word "and" but final

answer in a form such as $3 \leq k \leq -\frac{3}{4}$

scores A0

A1

4

[7]

Q19.

(a) $b^2 - 4ac = 144 - 4(k + 1)(k - 4)$

Clear attempt at $b^2 - 4ac$
Condone slip in one term of expression

M1

Real roots when $b^2 - 4ac \geq 0$

Not just a statement, must involve k

B1

$36 - (k^2 - 3k - 4) \geq 0$
$\Rightarrow k^2 - 3k - 40 \leq 0$

***AG** (watch signs carefully)*

A1

3

(b) $(k - 8)(k + 5)$

Factors attempt or formula

M1

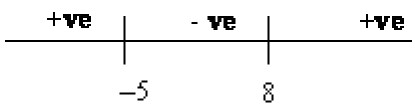
Critical points 8 and –5

A1

Sketch or sign diagram **correct,** must have 8 and –5

M1

$-5 \leq k \leq 8$



A1

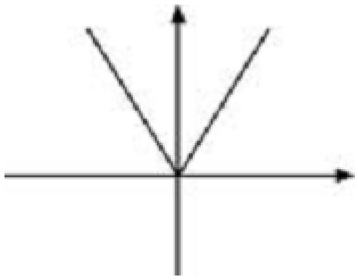
4

A0 for $-5 < k < 8$ or two separate inequalities unless word AND used

[7]

Q20.

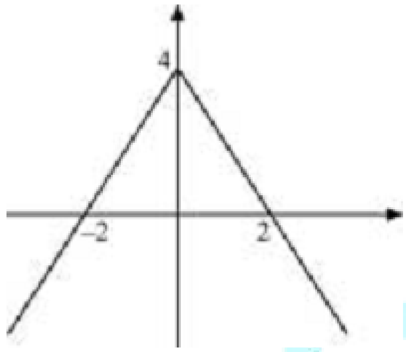
(a)



B1

1

(b)



Shape inverted V in all four quadrants

M1

Symmetrical about y-axis

A1

Coordinates

A1

3

(c) $4 - |2x| = x$

M1

$4 - 2x = x$ $\quad x = \frac{4}{3}$

Attempt to solve

A1

$4 + 2x = x$ $\quad x = -4$

And no others

A1

3

(d) $-4 < x < \frac{4}{3}$

Either correct

M1

Other solution and no extras

SC $-4 \le x \le \frac{4}{3}$ *B1*

A1

2

[9]

Q21.

(a) $b^2 - 4ac = 4 - 4(k-1)(2k-3)$

(or seen in formula) condone one slip

M1

Real roots when $b^2 - 4ac \ge 0$

must involve f(k) ≥ 0 (usually M1 must be earned)

E1

$4 - 4(2k^2 - 5k + 3) \ge 0$

$\Rightarrow -2k^2 + 5k - 3 + 1 \ge 0$

at least one step of working justifying ≤ 0

$\Rightarrow 2k^2 - 5k + 2 \le 0$

AG

A1

3

(b) (i) $(2k-1)(k-2)$

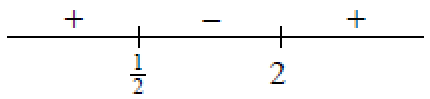
B1

1

(ii) (Critical values) $\frac{1}{2}$ and 2

ft their factors or correct values seen on diagram, sketch or inequality or stated

B1 √



use of sketch / sign diagram

M1

$\Rightarrow$ $0.5 \le k \le 2$

M1A0 for $0.5 < k < 2$ or $k \ge 0.5$, $k \le 2$

A1

3

[7]

Q22.

(a) $y_D = 3 + 1 = 4$ or $y_C = 12 - 8 = 4$

Attempt at either y coordinate

M1

Area $ABCD$ = $3 \times 4 = 12$

A1

2

(b) (i) $x^3 - \frac{x^4}{4}$ $(+C)$

Increase one power by 1

M1

One term correct unsimplified

A1

All correct unsimplified (condone no +C)

A1

3

(ii) Sub limits −1 and 2 into their (b)(i) ans

May use both −1, 0 and 0, 2 instead

M1

$[8 - 4] - \left[-1 - \frac{1}{4}\right]$ $= 5\frac{1}{4}$

A1

Shaded area = "their" (rectangle− integral)

Alt method: difference of two integrals

M1

$= 12 - 5\frac{1}{4} = 6\frac{3}{4}$

CSO. *Attempted M2, A2*

A1

4

(c) (i) $\frac{dy}{dx} = 6x - 3x^2$

One term correct

M1

All correct (no +C etc)

A1

2

(ii) When $x = 1$, $y = 2$ when $x = 1$,

May be implied by correct tgt equation

B1

$\frac{dy}{dx} = 3$ as 'their' grad of tgt

Ft their derivative when $x = 1$

M1ft

Tangent is $y - 2 = 3(x - 1)$

Any correct form $y = 3x - 1$ etc

A1

3

(iii) Decreasing when $\frac{dy}{dx} = 6x - 3x^2 < 0$

Watch no fudging here!!
May work backwards in proof.

M1

$3(2x - x^2) < 0 \quad \Rightarrow \quad x^2 - 2x > 0$

AG *(be convinced no step incorrect)*

A1

2

(d) Two critical points 0 and 2

Marked on diagram or in solution

M1

$x > 2$, $x < 0$ ONLY

or M1 A0 for $0 < x < 2$ or $0 > x > 2$

***SC** B1 for $x > 2$ (or $x < 0$)*

A1

2

[18]

Q23.

(a) $k^2 + 10k + 25 - 12k^2 - 24k$

Condone one slip

M1

$= -11k^2 - 14k + 25$

No ISW here

A1

2

(b) (i) Real roots when "$b^2 - 4ac$" ≥ 0

Non-negative discriminant (stated / used)

B1

$(k + 5)^2 - 12k(k + 2)$

Finding $b^2 - 4ac$ in terms of k
Or factorisation attempt.

M1

$(k - 1)(11k + 25)$ attempted to be shown

m1

equal to $11k^2 + 14k - 25$

$-11k^2 - 14k + 25 \geq 0$

$\Rightarrow (k - 1)(11k + 25) \leq 0$

Real roots condition correct and ...
AG (be convinced about inequality)

A1

5

(ii) (Critical values) 1 and $-\frac{25}{11}$ seen

B1

Sketch or sign diagram

M1

$\Rightarrow -\frac{25}{11} \le k \le 1$

A1

3

[10]

Q24.

(a) $3x - 3 > 3 - 5x - 30$

Multiplying out (condone one slip)

M1

$\Rightarrow 8x > -24$

Or correct equivalent eg $-8x < 24$

A1

$\Rightarrow x > -3$

(Penalise ≤ , ≥once only in (a) and (b))

A1

3

(b) $x^2 - x - 6 = (x - 3)(x + 2)$

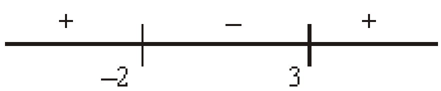
Attempt to use quad formula or factorise

M1

(critical points are) 3 and –2

May be seen in diagram or solution

A1



Sketch or sign diagram

M1

$2 < x < 3$

A1

4

[7]